

Count-controlled loops

Revise 360 lesson plan — 60 minutes

Unit	Beyond OCR J277: a Python pathway supporting 2.1, 2.2 and 2.3
Lesson	Count-controlled loops (lesson 4 of the unit)
Lesson ID	pr-104
Experience ID	pr-104
Plan ID	PLAN-PR-L04
Content version	03c522a8c36f
Website	https://revise360.co.uk/experience.html?id=pr-104

Big question, outcomes and success criteria

Big question — How do you decide what to put inside range() so that a loop counts exactly the values you want?

Prerequisites

The lesson before this one is pr-103. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: pr-101, pr-102.

Assessable outcomes

1. **pr-104-o1** — know how to write a for loop that repeats a block a set number of times
2. **pr-104-o2** — understand how range() decides which values the loop counter takes
3. **pr-104-o3** — know how to build a running total inside a loop, and how to put one loop inside another

Success criteria the pupil can check against

Taken from the briefs the independent activities themselves show, so these are the words the pupil sees. Their work has met the outcome when it does these things.

- Change only the number inside range(). **pr-104-s1-a3**
- Display exactly: ping **pr-104-s1-a3**
- Replace ____ with one instruction of your own. **pr-104-s1-a4**
- Keep it indented by four spaces. **pr-104-s1-a4**
- Display exactly: ready **pr-104-s1-a4**
- The user types in one word. **pr-104-s1-a5**
- Three lines of output, all the same. **pr-104-s1-a5**
- The word is only asked for once. **pr-104-s1-a5**

Vocabulary, resources and preparation

Key vocabulary — Count-controlled iteration, range(), Running total, Nested loop

Resources

SLIDES	presentations/PR_L04_CountControlledLoops_Lesson.pptx
WORKSHEET	worksheets/PR_L04_CountControlledLoops_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=pr-104

Before the lesson

1. Sign in on the **teacher dashboard** (`teacher.html`) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the Python runtime and the 360 image are already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on <code>PR_L04_CountControlledLoops_Worksheet.docx</code> is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong before moving on.	Correct their own starter.	A correction written, not just a tick.
8-16	Explain and model	The new idea, then the worked example, typed live rather than shown finished.	Watch the example being built, and say what each line does.	They can name the line that does the job, not just copy it.
16-19	Set up and settle	Sign-in, launch, and what to record on paper before anyone types.	Sign in, open the lesson, read the task before typing.	Everyone is at station 1 activity 1, not typing yet.
19-48	Online: work the stations in order	Pupils work; you circulate and read the editor over shoulders.	Work the stations in order, running and checking as they go.	The station badges fill in; the editor shows their own program, not the starter.
48-53	Check understanding	The hinge question, hands down, then the reteach if the class splits.	Answer the hinge question, hands down.	The split across the options, before any reteach.
53-58	Apply or extend	The challenge activity, or the exam practice on the worksheet.	The Challenge activity, or the exam practice.	worksheet, Exam practice a) to c)

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
58-60	Exit and set up the next lesson	The exit question and what to finish before next time.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Repeating a line

- `for i in range(4):` runs the indented block below it four times.
- Count-controlled iteration is used when the number of repeats is known in advance.
- Common mistake: leaving the line to repeat outside the loop, so it only runs once.

Challenge on this wall: How would you repeat a line as many times as the user asks for?

2. Using the counter

- The loop variable holds the count on each pass, so you can print it or calculate with it.
- `range(n)` counts from 0, so the values are 0 up to $n - 1$.
- Common mistake: expecting `range(5)` to include 5, or to start at 1.

Challenge on this wall: What has to change so that a loop counts from 1 up to and including n ?

3. Choosing the range

- `range(start, stop)` counts from start up to but not including stop.
- A third value is the step, so `range(10, 0, -2)` counts back down in twos.
- Common mistake: forgetting stop is never reached, which quietly loses the last value.

Challenge on this wall: Which range counts through 4 to 9, with both of them included?

4. Running totals

- A running total is set to 0 before the loop and grows inside it.
- `total = total + value` adds this pass's value to whatever is already there.
- Common mistake: setting the total back to 0 inside the loop, which loses everything counted so far.

Challenge on this wall: Where must a running total be set to zero, and why there?

5. Loops inside loops

- A nested loop runs the whole inner loop on every single pass of the outer one.
- Rows and columns are the usual job: the outer loop picks the row, the inner one the column.
- Common mistake: indenting the inner loop's body wrongly, so all the rows run together.

Challenge on this wall: How many lines does a loop of 4 columns inside a loop of 3 rows print?

6. Putting it together

- Decide what each loop counts before you write it, and name the loop variable after that.
- Build the total, the count and the average from the same single pass through the data.
- Common mistake: printing inside the loop when the task asks for one line at the end.

Challenge on this wall: Which figures can one pass through the data work out at the same time?

The example to model

A loop repeats instructions. `for i in range(3):` means take three turns. The line underneath is indented by four spaces, and that indent is what says the line belongs to the loop. Note the colon at the end of the `for` line.

```
for i in range(3):
    print("tick")
```

It displays **tick / tick / tick**

Decisions to model out loud, not just the finished answer: say why you chose that name, why that line comes before the next one, and what you would try first if it did not work. The example deliberately uses different data from the task, so a pupil cannot copy it and be finished.

Retrieval and the check for understanding

Starter — Write down every number printed by for i in range(2, 9, 3), then say how many lines of output there are altogether.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pr-104-s3-c6), so the wording a pupil sees matches the wording you say.

What will this program display?

```
for n in range(1, 10, 3):
    print(n)
```

Correct: 1 4 7

What a pupil might choose instead

- 1 4 7 10
- 1 2 3

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: The step of 3 jumps from 1 to 4 to 7. The next jump would be 10, which is the stop, so the loop ends.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Misconceptions this lesson addresses

- leaving the line to repeat outside the loop, so it only runs once. [wall panel](#)
- Python is complaining that it expected something after the colon. [pr-104-s1-a8, a Fix it activity](#)
- expecting range(5) to include 5, or to start at 1. [wall panel](#)
- One line inside the loop is changing the counter as well as range(). [pr-104-s2-a8, a Fix it activity](#)
- forgetting stop is never reached, which quietly loses the last value. [wall panel](#)
- Count the lines that come out and compare it with the numbers you typed. [pr-104-s3-a8, a Fix it activity](#)
- setting the total back to 0 inside the loop, which loses everything counted so far. [wall panel](#)
- Something inside the loop wipes the total out before it has a chance to grow. [pr-104-s4-a8, a Fix it activity](#)

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	17	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	13	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.
Produce	19	The pupil writes it from the brief and nothing else.

Challenge 6 The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Repeating a line	9	Iteration is the proper name for repetition, and it is one of the three basic...
2	Using the counter	9	Counting from 0 is normal in programming, and it is why the first item of a list is item...
3	Choosing the range	9	The stop value of a range is the one value the loop never...
4	Running totals	9	A counter and a running total work the same way: one adds 1 and the other adds...
5	Loops inside loops	9	Nesting one loop inside another multiplies the work rather than...
6	Putting it together	9	One pass that gathers several answers at once beats looping over the same data...
final	Final challenge: how many times?	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Python: the order is not negotiable. Each activity teaches what the next one needs, so a pupil works through them in order and completes every one. A shorter lesson, homework or a slow worker does not authorise skipping: the pupil stops where they are and resumes at their earliest unfinished question next time, because progress saves after every answer.

A paper activity, a help request or a tick on this plan does not mark an online question complete, and there is no teacher bypass. If a pupil is stuck after the hint, that is the moment to teach them, not to move them on.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pr-104-x1	A program should print the numbers 1 to 10 but prints 0 to 9 instead. Its loop begins <code>for i in range(10)</code> . State which values the original <code>range()</code> counted through, and rewrite the line so the loop prints 1 to 10.	2	worksheet, Exam practice a); deck, Exam practice slide
pr-104-x2	Write a program that reads how many numbers there are, then reads that many whole numbers and prints their total and their average.	5	worksheet, Exam practice b); deck, Exam practice slide
pr-104-x3	Write a program that reads a whole number and prints that many lines, the first holding one * character, the second two, and so on.	4	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pr-104.json`, which is not published with the website.

Every programming activity in this lesson is marked by running the pupil's program against its test cases, so the mark is the program's behaviour and not its appearance. The model solutions are in `answers/codebank/pr-104.json`.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What will this program display?** — start beep end; start end beep beep
- **What will this program display?** — in in out out; in out
- **What will this program display?** — 1 2 3; 0 1 2 3
- **What will this program display?** — 0 1 2 3 4; 2 3 4 5 6
- **What will this program display?** — 2 3 4 5 6; 3 4 5

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- Writing a for loop with the right range
- Keeping a running total inside a loop

- Using one loop inside another

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is pr-105, Condition-controlled loops. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

If this lesson takes more than an hour

It will. There are 55 activities across 6 stations, and a beginner does not finish 55 of them in twenty-nine minutes. Plan it as two or three sessions of an hour: stations 1 and 2 in the first, 3 and 4 in the second, 5, 6 and the final challenge in the third. Each session keeps the same starter-explain-model opening; the online block resumes where the class got to. Honest total: two to three hours.

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

A shorter lesson still does not authorise skipping a question. The save-and-resume point is wherever the pupil got to; there is no way to mark a question complete without completing it, and no teacher bypass.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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